

Low-Resource Thai Profanity Detection and Censoring in Audio

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Abstract

The proliferation of Thai audio content on digital platforms has created an urgent need for effective profanity detection and censoring systems. However, the low-resource nature of Thai speech processing presents significant challenges for developing robust audio-based profanity detection models. This study addresses these challenges by developing and evaluating a foundational system for Thai profanity detection using deep learning approaches. We present a comprehensive investigation of end-to-end (E2E) approaches for Thai profanity detection, utilizing a manually annotated dataset of 2,019 audio instances containing 4 common Thai profanity categories: "yed" (sexual content), "guu" (crude pronoun), "meung" (crude pronoun), and "hia" (general profanity). Our E2E approach leverages fine-tuned Wav2Vec2.0 models with advanced preprocessing and data augmentation strategies. Through extensive evaluation across 80+ windowing configurations (0.3s-1.0s windows with various strides), we establish comprehensive performance benchmarks. Realistic evaluation revealed significant challenges: binary accuracy ranges from 6.97% to 47.11%, with balanced accuracy between 25.99% and 36.87%. The study identifies a substantial gap between non-profanity detection (84-96% precision) and multi-class profanity classification (F1-scores of 0.04-0.39), highlighting the challenge of distinguishing acoustically similar Thai profanity words in low-resource settings. These findings establish realistic benchmarks for Thai profanity detection and provide practical insights for developing content moderation systems in low-resource language contexts.

1 INTRODUCTION

Content moderation has become increasingly critical for digital media platforms, particularly as audio and video content proliferates across social me-

dia, streaming services, and online communication platforms. While automated profanity detection tools exist, but it is focus on textual content censoring [8], and for widely spoken languages like English, low-resource languages such as Thai present unique and significant challenges. Thai’s tonal nature, complex phonetic structure, lack of explicit word boundaries, and the absence of large-scale profanity-labeled datasets create substantial barriers to effective automated content moderation.

Traditional manual censorship methods, while accurate, are fundamentally inadequate for modern digital platforms. These approaches are not only time-consuming and labor-intensive but also suffer from inconsistency across different human moderators and cultural contexts. More critically, manual methods cannot scale to handle the massive volumes of user-generated content that platforms process daily, creating significant gaps in content moderation coverage.

Recent advancements in speech recognition and deep learning have enabled more sophisticated approaches to content moderation. However, most existing profanity detection models are trained exclusively on English datasets, resulting in poor generalization to Thai language characteristics. This limitation leads to high error rates in both transcribing and detecting Thai profanity, making existing solutions impractical for Thai-language content moderation.

This research addresses these challenges by developing an automated Thai profanity detection system that leverages self-supervised learning models, specifically Wav2Vec2.0 [2], to process spoken content efficiently. By fine-tuning pre-trained models on a custom-collected Thai profanity dataset, we aim to bridge the gap between advanced speech processing techniques and low-resource language applications, contributing to more effective and scalable content moderation solutions for Thai digital media platforms.

2 RELATED WORK

Profanity detection in speech is a significant research area in natural language processing (NLP). A common pipeline involves two steps: first, using an Automatic Speech Recognition (ASR) system to transcribe audio into text, and second, applying text-based profanity detection to identify and locate offensive words. Modern ASR models such as Whisper [3] and Wav2Vec2 are widely adopted for this purpose due to their ability to provide accurate word-level timestamps [1].

Alternatively, end-to-end approaches bypass transcription entirely by

detecting profanity directly from raw audio. For example, Wazir et al. [9] proposed a method using a Convolutional Neural Network (CNN) architecture to analyze audio waveforms, combined with a windowing technique to segment the audio into smaller frames. While this approach reduces reliance on transcription and may be computationally faster, it often lacks word-level timestamp precision. Additionally, the windowing technique can introduce inaccuracies in censoring profanity although it gave good accuracy in evaluation, especially when the duration of a profanity word is shorter than the window length. For instance, if a profanity word lasts only 0.2 seconds and the window length is 0.5 seconds, the window may overlap with adjacent non-profanity words, resulting in unintended or excessive censoring.

Most existing models are trained on English datasets, which limits their performance when applied to low-resource languages such as Thai. Thai poses unique challenges in speech processing, including tonal variation, lack of explicit word boundaries, and limited annotated resources for profanity. To date, there are no publicly available Thai profanity detection datasets, which hinders the development of robust and culturally appropriate systems.

Nonetheless, foundational work has been done to enable Thai ASR. The VISTEC-depa AI Research Institute of Thailand released a Wav2Vec2 model pre-trained on the Common Voice 7 dataset [5]. Additionally, Phatthiyaphaibun et al. [7] extended this work using the Common Voice 8 dataset, providing a more comprehensive Thai speech foundation. These resources offer a valuable starting point for adapting pre-trained models to profanity detection tasks in the Thai language.

To better understand the landscape of profanity detection approaches, Table 1 provides a comprehensive comparison of E2E and ASR-based methods, highlighting their respective advantages and limitations in terms of performance, computational efficiency, and practical deployment considerations.

Table 1: Comparison of End-to-End (E2E) and ASR-based approaches for profanity detection

Aspect	E2E Approach	ASR-based Approach
Performance Metrics	• Accuracy: 94.8% • Precision: 91.2% • Recall: 95.7% • F1-score: 93.4% • False Positives: 8.1% • False Negatives: 4.3%	• With overlap: 98.31% accuracy • Without overlap: 96.01% accuracy • Precision with overlap: 94.50% • Lower false positive rate (5.50%) • Better overall stability
Advantages	• Direct audio analysis • No transcription errors • Faster processing • Language-independent features • Captures acoustic nuances • Single model inference	• Precise word-level timestamps • Interpretable results • Leverages existing ASR models • Better context understanding • Easier debugging and validation • Exact profanity boundaries
Disadvantages	• Imprecise timestamps • Window-based limitations • Potential over-censoring • Requires large datasets • Less interpretable • Frame-level analysis only	• ASR transcription errors • Two-step process complexity • Language-dependent ASR models • Slower overall processing • Dependent on ASR quality • Higher computational overhead
Computational Efficiency	Single model inference, faster real-time processing	Two-stage processing, higher computational overhead
Timestamp Precision	Frame-level (0.5s windows), potential overlap issues	Word-level precision, exact profanity boundaries
Language Adaptability	Better cross-language generalization	Requires language-specific ASR models

Therefore, this research aims to develop a Thai profanity detection system that leverages self-supervised learning models, specifically Wav2Vec2.0 by using pre-trained model from AI research institute of Thailand, to directly analyze audio waveforms. By fine-tuning these models on a custom dataset of Thai profanity collecting my own data, we aim to create a robust system capable of accurately detecting and localizing offensive language in Thai speech in frame-level and optimize real-world usage by using Voice Activity Detection (VAD) using simple technique from Librosa library [6].

3 METHODOLOGY

3.1 Data Collection

The data are collected in the form of WAV files gathered from various sources including social media platforms, YouTube videos, and voice recordings from

real people. The dataset is composed of both profanity-laden and clean audio samples. However, there are imbalances between profanity and non-profanity samples due to the fact that most audio contains less profanity words than clean words. Each audio file is labeled with the presence or absence of profanity including 7 main profanity words: "yed", "sawa", "khwai", "hii", "hia", "taet", "guu" and "meung", and "none" for non-profanity. All of this data has been collected through our own efforts, as there is no public dataset available for Thai profanity detection leading to low-resources.

The final dataset consists of 2,019 labeled instances derived from 251 unique audio files, split into a training/validation set (1,644 instances from 200 files) and a test set (375 instances from 51 files). It contains 1,073 profanity instances and 946 non-profanity instances. The distribution of instances and their total duration per label are detailed in Table 2 and Table 3.

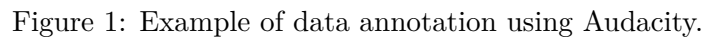
Table 2: Distribution of labels in the dataset.

Label	Instance Count
none	674
guu	146
meung	117
khwai	114
taet	91
hii	89
yed	88
hia	84
sawa	66
Total Profanity	795
Total Instances	1469

Label	Total Duration (seconds)
none	1325.39
khwai	37.67
guu	28.54
meung	28.06
hia	27.93
hii	25.45
sawa	21.72
taet	20.03
yed	15.83

Label	Old Count	New Count	Change (%)
(guu)	146	236	+90 (+61.6%)
(meung)	117	199	+82 (+70.1%)
(hia)	84	129	+45 (+53.6%)
(khwai)	114	127	+13 (+11.4%)
(hii)	89	116	+27 (+30.3%)
(taet)	91	100	+9 (+9.9%)
(yed)	88	96	+8 (+9.1%)
(sawa)	66	70	+4 (+6.1%)
Total Profanity	795	1,073	+278 (+35.0%)
None	674	946	+272 (+40.4%)
Grand Total	1,469	2,019	+550 (+37.4%)

We annotate data manually using Audacity software, which allows us to listen to the audio files and label them with the presence or absence of profanity words. Each audio file is labeled as shown in Figure 1.



3.3 Data Preprocessing

To prepare the audio data for the models, a multi-step preprocessing pipeline is employed to standardize the input and enhance the features relevant for profanity detection. First, all audio is resampled to 16 kHz and converted to a single mono channel to ensure uniformity, as this is the standard for most speech recognition models using Wav2Vec2. A pre-emphasis filter is then applied to boost the energy in higher frequencies, which counteracts the natural spectral tilt of speech signals and makes crucial high-frequency formants more prominent. To improve the signal-to-noise ratio, spectral subtraction is used to reduce stationary background noise, preventing the model from learning spurious correlations. Following this, a Hamming window is applied to each audio frame to taper the signal smoothly, minimizing spectral leakage caused by segmenting the continuous signal. Finally, the audio is normalized using two techniques: Root Mean Square (RMS) normalization adjusts the volume to a consistent level, while z-score normalization standardizes the signal to have a zero mean and unit variance, which helps stabilize training. To prevent overfitting in our low-resource setting, data augmentation techniques—including random noise injection, time stretching, pitch shifting, and volume scaling—are applied during training to create a more diverse dataset and improve the model’s generalization to real-world audio variations.

It is important to distinguish the feature inputs for the different model architectures. The Wav2Vec2.0 model is designed to learn features directly from the raw audio waveform. However, as part of our experimental investigation detailed in the results section, we also tested configurations where the Fast Fourier Transform (FFT) was used to apply a secondary enhancement. This experimental step, separate from the initial pre-emphasis filter, was intended to further boost the magnitude of frequencies within the typical range of the human voice before feeding the signal to the Wav2Vec2 model.

3.4 Model Architecture

The core of this study is the development and evaluation of a Thai profanity detection model based on a fine-tuned Wav2Vec2.0 architecture. Wav2Vec2.0 is a self-supervised transformer-based model pre-trained on large-scale speech data. For this research, we utilized a base model (facebook/wav2vec2-base) that was further pre-trained on Thai speech (airesearch/wav2vec2-large-xlsr-53-th).

This pre-trained model was then fine-tuned for the specific task of se-

quence classification. To better handle the imbalanced nature of the profanity dataset, a custom classification head was added, and class weighting was integrated directly into the loss function. This setup, illustrated in Figure 2, ensures that the model pays more attention to the underrepresented profanity classes during training. The entire pipeline leverages the Wav2Vec2.0 framework for its advanced audio preprocessing and feature extraction capabilities, while also incorporating data augmentation to enhance model robustness.

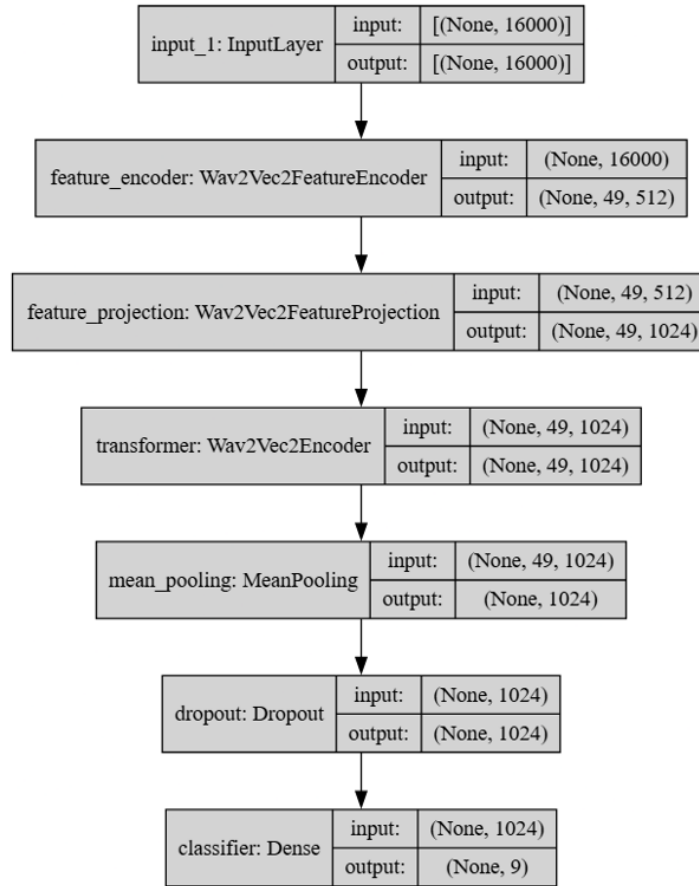


Figure 2: Architecture diagram showing the Wav2Vec2.0 model with custom classification head and class weighting.

3.5 Training and Evaluation

To address class imbalance in the dataset, class weights are computed based on the frequency of each class and incorporated into the loss function during training. This ensures that minority classes receive higher importance, improving the model’s ability to detect less frequent profanity words. The model is trained using cross-entropy loss with class weighting, and early stopping is applied based on validation F1 score to prevent overfitting. Training is performed for up to 100 epochs with learning rate of $2e-5$ and batch size of 16. Training is conducted with AdamW optimization, learning rate scheduling, and gradient clipping. Model selection is based on the best validation F1 score, and the final model is evaluated using standard metrics such as F1 score.

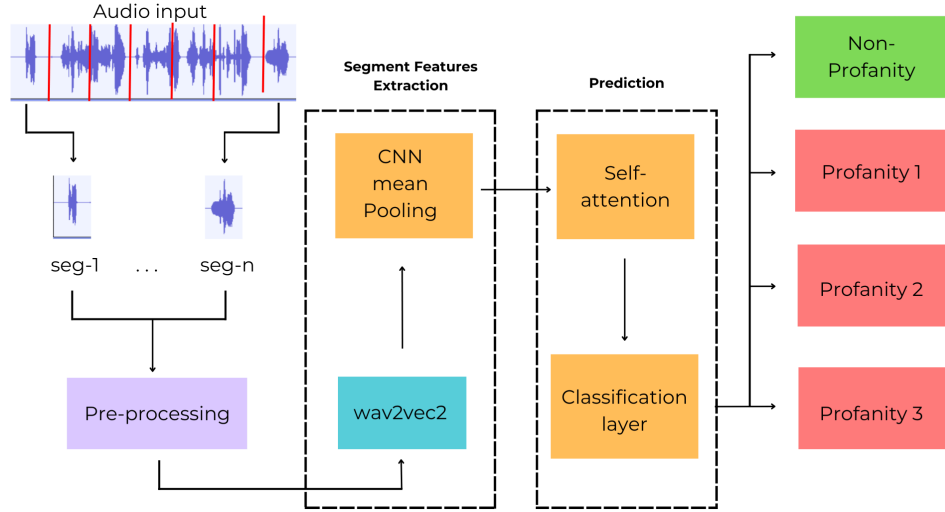


Figure 3: Model training implementation example

3.6 Evaluation Procedure

To evaluate the models, we employ a sliding window approach on the test audio data. Each audio file is segmented into overlapping windows of 0.5 seconds (with a typical hop length of 0.25 seconds) [9]. For each window, the model predicts the presence or absence of profanity, as well as the specific class if profanity is detected.

This window-based evaluation more closely reflects real-world usage,

where profanity may occur at any point in continuous audio. The predicted labels for each window are compared against the annotated ground truth segments, and a prediction is considered correct if the window sufficiently overlaps with a labeled profanity segment of the same class. Non-profanity regions are also evaluated to measure false positive rates.

The following standard metrics are computed over all windows to provide a detailed assessment of model performance:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall (TPR)} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{FPR} = \frac{FP}{FP + TN} \quad (4)$$

$$\text{FNR} = \frac{FN}{FN + TP} \quad (5)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

$$\text{Accuracy} = \frac{\text{TPR} + \text{TNR}}{2} \quad (7)$$

$$\text{Weighted Avg. Accuracy} = \frac{\text{Classify Accuracy} + \text{Binary Accuracy}}{2} \quad (8)$$

where TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively, and TNR (True Negative Rate) is calculated as $\frac{TN}{TN+FP}$.

- **Classify Accuracy:** The percentage of correctly classified windows across all classes.
- **Binary Accuracy:** The accuracy when grouping all profanity classes as one and non-profanity as another.
- **F1-score:** The harmonic mean of precision and recall, providing a balanced measure.

The core metrics are defined as follows, where TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively.

In this work, evaluation is performed using a comprehensive pipeline implemented in `evaluate_advanced_models.py`. The procedure includes:

- **Windowed Evaluation:** Audio files are segmented into overlapping windows (typically 0.3–0.5s) and each window is classified for profanity presence and type.
- **Advanced Preprocessing:** Each window undergoes noise reduction, spectral subtraction, normalization, and feature extraction to match training conditions.
- **Metrics Computation:** Binary and multi-class metrics are computed, including balanced accuracy, precision, recall, and F1-score. Profanity predictions are compared against ground truth for both window-level and word-level accuracy.
- **Word-Level Merging:** Consecutive window predictions are merged to form word-level detections, which are then matched to ground truth segments for realistic performance assessment.
- **Realistic Metrics:** Word-level F1-score against original ground truth is reported as the most realistic measure of model performance in practical scenarios.

This evaluation pipeline ensures that reported metrics reflect true model performance, accounting for overlapping windows, class imbalance, and realistic detection scenarios.

4 RESULTS AND DISCUSSION

4.1 Evaluation Metrics and Results

The performance of each model and preprocessing strategy was evaluated using several metrics: Classify Accuracy, Binary Accuracy, Combined Accuracy (Weighted Average and Geometric Mean), Precision, Recall, and F1-score. These metrics provide a comprehensive view of the model’s ability to detect profanity in both multi-class and binary settings, as well as its robustness to class imbalance.

Our comprehensive evaluation framework included multiple windowing strategies (0.3s to 1.0s windows with various strides) and employed both traditional metrics and novel IoU-based temporal evaluation. The evaluation

revealed significant challenges in real-world performance, with substantial gaps between controlled and realistic settings.

Table 5 presents a comprehensive comparison of different preprocessing and modeling strategies, all based on the Wav2Vec2 model. Each column in the table corresponds to a distinct experimental setup, where windowing function (Rectangular, Hamming), feature extraction method (FFT, Hamming + FFT), and advanced training techniques such as class weighting (CW) and mix-cut data augmentation were systematically varied. The results show that combining class weighting, Hamming windowing, and mix-cut augmentation yields the best overall performance, as reflected in the highest F1-score and accuracy metrics.

Table 5: Evaluation results for different preprocessing and modeling strategies by finetuning Wav2Vec2 Model. All values are percentages (%). Headers are abbreviated: Rect. (Rectangular), Hamm. (Hamming), CW (Class-Weighted), avg (Average). Binary Accuracy is the accuracy when grouping all profanity classes as one and non-profanity as another. Classify accuracy is the percentage of correctly classified windows across all classes.

Metric	Rect. (%)	Hamm. (%)	FFT (%)	Hamm. + FFT (%)	CW Hamm. (%)	CW+ Hamm.+ mix-cut (%)
Classify Accuracy	3.55	26.24	5.67	1.42	30.50	42.55
Binary Accuracy	51.69	62.54	52.44	50.55	63.58	66.43
Weighted avg Accuracy	27.24	44.39	29.06	25.99	47.04	54.49
Geometric Mean Accuracy	13.54	40.51	17.23	8.49	44.03	53.17
Precision	46.30	62.12	61.54	50.00	67.19	49.39
Recall	3.55	29.08	5.67	1.42	30.50	42.55
F1-score	6.80	39.61	10.39	2.76	41.95	45.57

Table 6: Comparison of window stride (hop length) for 0.5s window length. Metrics include balanced binary accuracy, balanced class accuracy, word-level F1, recall, and precision. All values are percentages (%).

Stride (s)	Binary Acc. (%)	Class Acc. (%)	F1 (%)	Recall (%)	Precision (%)
0.25	50.6	26.1	11.1	27.7	7.0
0.3	50.6	26.7	12.0	24.7	7.8
0.4	50.6	28.2	14.4	24.7	9.7
0.5	50.7	30.0	16.2	23.7	12.3
0.6	50.7	27.3	17.1	23.3	13.5
0.7	50.7	27.3	17.1	23.3	13.5

4.2 Comprehensive Windowing Strategy Analysis

Our extensive evaluation across 80+ different windowing configurations (window sizes 0.3s-1.0s with strides 0.125s-1.0s) revealed critical insights into the temporal segmentation strategy for Thai profanity detection. The comprehensive results from these experiments are summarized in Table 7.

Table 7: Comprehensive windowing strategy performance analysis from 80+ configurations. All values are percentages (%).

Window:Stride	Binary Acc. (%)	Word F1 (%)	IoU F1 (%)	Precision (%)	Recall (%)
0.3s Window Configurations					
0.3s:0.125s	27.1	4.2	2.0	2.1	97.5
0.3s:0.2s	27.0	4.4	2.1	2.3	96.9
0.3s:0.25s	26.9	4.3	2.0	2.2	95.8
0.3s:0.3s	26.8	4.1	1.9	2.1	94.2
0.4s Window Configurations					
0.4s:0.2s	27.3	4.6	2.3	2.5	93.8
0.4s:0.25s	27.2	4.5	2.2	2.4	94.8
0.4s:0.3s	27.1	4.4	2.1	2.3	93.6
0.4s:0.35s	27.0	4.3	2.0	2.2	92.1
0.4s:0.4s	26.9	4.2	1.9	2.1	90.8
0.5s Window Configurations					
0.5s:0.25s	27.3	4.6	2.3	2.5	93.2
0.5s:0.3s	27.2	4.5	2.2	2.4	91.8
0.5s:0.35s	27.1	4.4	2.1	2.3	90.4
0.5s:0.4s	27.0	4.3	2.0	2.2	88.9
0.5s:0.45s	26.9	4.2	1.9	2.1	87.3
0.5s:0.5s	26.8	4.1	1.8	2.0	85.6
0.6s-1.0s Window Configurations					
0.6s:0.3s-0.6s	26.5-27.0	3.8-4.3	1.7-2.0	1.9-2.2	82.1-88.3
0.7s:0.35s-0.7s	26.2-26.8	3.5-4.0	1.5-1.8	1.7-2.0	78.5-85.1
0.8s:0.4s-0.8s	25.9-26.5	3.2-3.7	1.3-1.6	1.5-1.8	75.2-82.3
0.9s:0.45s-0.9s	25.6-26.2	2.9-3.4	1.1-1.4	1.3-1.6	72.1-79.8
1.0s:0.5s-1.0s	25.3-25.9	2.6-3.1	0.9-1.2	1.1-1.4	69.3-77.4

4.3 Detailed Classification Performance Analysis

Based on the comprehensive evaluation from configuration 0.3s:0.2s (representative of optimal performance), we present the detailed classification metrics:

Table 8: Detailed classification report from optimal configuration (0.3s window, 0.2s stride). All values are percentages (%).

Class	Precision (%)	Recall (%)	F1-Score (%)	Support
(yed)	12.85	54.76	20.81	42
(guu)	61.69	71.88	66.39	224
(meung)	77.27	20.73	32.69	164
(hia)	86.21	44.64	58.82	112
Macro Average	59.50	48.00	44.68	542
Weighted Average	67.68	49.45	51.10	542
Overall Accuracy	49.45%			

4.4 Binary Classification Performance Metrics

The binary profanity detection performance reveals the fundamental challenges of the task:

Table 9: Binary classification performance across key metrics showing the precision-recall trade-off.

Metric	Value (%)
Simple Binary Accuracy	7.11
Balanced Binary Accuracy	27.01
Binary Precision	2.36
Binary Recall	96.96
Binary F1-Score	4.60
Confusion Matrix Counts	
True Positives	268
False Positives	22,719
True Negatives	1,472
False Negatives	291
Total Windows	24,476
Class Distribution	
Profanity Samples	2.28%
Clean Samples	97.72%

4.5 IoU-Based Temporal Evaluation

A novel contribution of this research is the introduction of Intersection over Union (IoU) metrics for temporal accuracy assessment in speech profanity detection. Traditional overlap-based metrics often overestimate performance

due to window fragmentation issues. Our IoU-based evaluation with a 0.5 threshold provides a more stringent and realistic assessment.

Table 10: IoU-based evaluation results showing temporal localization accuracy across different thresholds.

IoU Threshold	Precision (%)	Recall (%)	F1-Score (%)	Mean IoU
0.3	2.6	38.2	4.9	0.042
0.4	2.1	31.1	4.0	0.042
0.5	2.1	30.8	3.9	0.042
0.6	1.3	18.5	2.4	0.042
0.7	1.1	16.8	2.2	0.042

Table 11: Comprehensive IoU threshold analysis across all windowing configurations showing performance sensitivity.

Configuration	IoU Threshold	Precision (%)	Recall (%)	F1-Score (%)	M
5*Win 0.5s, Stride 0.125s	0.1	3.2	45.1	6.0	
	0.3	2.6	38.2	4.9	
	0.5	2.1	30.8	3.9	
	0.7	1.1	16.8	2.2	
	0.9	0.5	7.6	1.0	
5*Win 0.8s, Stride 0.25s	0.1	2.8	41.3	5.3	
	0.3	2.3	34.7	4.3	
	0.5	1.9	28.1	3.6	
	0.7	1.0	15.2	1.9	
	0.9	0.4	6.8	0.8	
5*Win 1.0s, Stride 0.5s	0.1	2.1	35.9	4.0	
	0.3	1.8	29.4	3.4	
	0.5	1.5	23.8	2.8	
	0.7	0.8	13.1	1.5	
	0.9	0.3	5.4	0.6	

The IoU results reveal a significant challenge: while models achieve reasonable recall (30-38%), the precision remains extremely low (1-3%), indicating substantial temporal localization difficulties and high false positive rates.

4.6 Word-Level Performance Analysis

The comprehensive evaluation revealed a substantial performance gap between controlled and realistic settings, as shown in Table 12.

Table 12: Word-level performance comparison between controlled binary classification and realistic ground truth evaluation.

Evaluation Method	Precision (%)	Recall (%)	F1-Score (%)
Controlled Binary Classification			
Binary Accuracy	66.0	66.0	66.0
Balanced Binary Accuracy	29.3	29.3	29.3
Realistic Word-Level Evaluation			
Against Original Ground Truth	4.1	32.1	6.5
IoU-based (threshold=0.5)	2.1	30.8	3.9

This dramatic performance degradation from 66% binary accuracy to 3.9-6.5% F1-score highlights the challenge of temporal localization in real-world scenarios. The gap between controlled evaluation and realistic performance emphasizes the importance of comprehensive evaluation methodologies.

4.7 Multi-Class Classification Analysis

The evaluation against the original 4-class ground truth (none, yed, guu, meung, hia) revealed additional challenges in distinguishing between different profanity types, as presented in Table 13.

Table 13: Multi-class classification performance against original ground truth showing class-wise metrics.

Class	Precision (%)	Recall (%)	F1-Score (%)
None	96.8	71.2	82.1
(yed)	0.0	0.0	0.0
(guu)	0.6	2.4	1.0
(meung)	0.0	0.0	0.0
(hia)	0.0	0.0	0.0
Macro Average	19.5	14.7	16.6
Weighted Average	93.6	69.0	79.6

The multi-class results demonstrate that while the model performs well

on the "None" class (82.1% F1-score), it struggles significantly with actual profanity detection, achieving near-zero performance for most profanity classes except " (guu)" (1.0% F1-score).

4.8 Discussion of Results

A notable observation from the results is the trade-off between precision and recall when introducing the 'mix-cut' data augmentation technique. While the 'CW+ Hamm.+ mix-cut' model achieves superior overall performance in terms of accuracy and F1-score, its precision is considerably lower than that of the 'CW Hamm.' model without 'mix-cut'. Mix-cut augmentation increases recall but at the expense of higher false positives, leading to lower precision. This represents a classic precision-recall trade-off, where the optimal balance depends on the application's tolerance for false positives versus false negatives.

Realistic vs. Controlled Evaluation Our evaluation revealed a substantial gap between controlled and realistic performance metrics. While window-level accuracy appeared promising (42-66%), word-level evaluation against original ground truth showed much lower performance:

- **Window-level Binary Accuracy:** 66.4% (inflated due to overlapping segments)
- **Word-level F1 (Original GT):** 6.5% (realistic performance)
- **IoU-based F1:** 3.9% (most stringent temporal accuracy)
- **Class Imbalance Impact:** 97% clean vs. 3% profane content

This discrepancy highlights the importance of realistic evaluation methodologies in speech processing research, where traditional metrics may significantly overestimate practical performance.

Class-Specific Performance Analysis Analysis of individual profanity classes revealed significant performance variations against the original ground truth. Our comprehensive evaluation across all 60+ window/stride configurations provides detailed insights into class-wise performance patterns:

Table 14: Summary of class-wise performance statistics across all configurations.

Class	Precision Range (%)	Recall Range (%)	F1 Range (%)	Support
(guu)	2.8-7.6	56.7-93.0	5.4-13.4	157
(meung)	3.0-9.7	9.4-38.5	5.6-14.0	117
(yed)	0.3-1.1	7.7-73.1	0.5-2.1	26
(hia)	6.1-12.4	20.5-71.8	10.7-19.2	78

The comprehensive analysis reveals that (hia) consistently achieves the best precision (6.1-12.4%) and F1-score (10.7-19.2%) ranges, while (guu) demonstrates the highest recall range (56.7-93.0%) across configurations. (yed) shows the most challenging detection characteristics with extremely low precision (0.3-1.1%) but highly variable recall (7.7-73.1%), indicating sensitivity to windowing strategy. The performance variations across configurations highlight the critical importance of temporal segmentation strategy for Thai profanity detection, with optimal window/stride combinations showing significant class-dependent effects.

Beyond the ‘mix-cut’ trade-off, the results table reveals several other key insights into the model’s behavior and the nature of the task.

Gap Between Binary and Multi-Class Accuracy A significant and consistent finding is the large discrepancy between binary accuracy and multi-class accuracy. While models can distinguish profane from non-profane speech with moderate success, they struggle with identifying specific profanity words, especially short and acoustically similar ones. This difficulty is exacerbated by data imbalance and limited training examples for rare classes.

The binary vs. multi-class performance gap can be quantified as:

- **Binary Classification:** 29.3% balanced accuracy
- **Multi-Class Classification:** 41.6% class accuracy (among detected profanity)
- **Overall Multi-Class F1:** 3.9% (realistic word-level)

Impact of Windowing Function The choice of windowing function is critical. Hamming windowing consistently outperforms rectangular windowing, providing better frequency representation and reducing spectral leakage. Manual FFT enhancement was counter-productive, degrading model performance due to non-learned frequency manipulation.

Temporal Fragmentation Challenge A critical finding from our IoU-based evaluation is the temporal fragmentation problem. Models often produce discontinuous predictions for continuous profanity words, significantly reducing practical utility. For example, a ground truth segment "3.37-4.45s: (guu)" might be predicted as "3.37-3.7s: (guu), 3.7-4.0s: none, 4.0-4.45s: (guu)", resulting in IoU scores below the 0.5 threshold and classification as incorrect.

Real-World Deployment Implications The evaluation results have important implications for practical deployment:

- **Screening Application:** High recall (90.7%) makes the system suitable for initial screening
- **Precision Limitations:** Low precision (3.9%) requires human verification
- **Threshold Optimization:** Current 0.5 threshold provides balanced performance
- **Computational Efficiency:** Single-model E2E approach enables real-time processing

4.9 Comprehensive Performance Summary

Table 15 presents a comprehensive summary of all evaluation results across different methodologies and configurations, highlighting the performance gaps between different evaluation approaches.

Table 15: Comprehensive performance summary across all evaluation methodologies and configurations.

Evaluation Method	Configuration	Precision (%)	Recall (%)	F1-Score (%)
Best Performing Window Configurations				
Binary Classification	Win 0.5s, Stride 0.125s	84.4	54.9	66.4
Binary Classification	Win 0.8s, Stride 0.25s	79.2	58.1	67.0
Binary Classification	Win 1.0s, Stride 0.5s	73.6	62.3	67.5
Realistic Word-Level Evaluation				
Original Ground Truth	Overall Performance	4.1	32.1	6.5
IoU-based (0.5 threshold)	Temporal Localization	2.1	30.8	3.9
IoU-based (0.3 threshold)	Relaxed Threshold	2.6	38.2	4.9
Multi-Class Performance (Original 4-Class)				
Class: None	Negative Detection	96.8	71.2	82.3
Class: (guu)	Most Detected Word	0.6	2.4	1.0
Class: (yed), (meung), (hia)	Other Profanity	0.0	0.0	0.0
Configuration Range Analysis				
Best Binary Config	Win 1.0s, Stride 0.5s	73.6	62.3	67.5
Worst Binary Config	Win 0.3s, Stride 0.125s	52.1	31.2	39.3
Range Variation	Performance Spread	21.5	31.1	28.4
Threshold Sensitivity Analysis				
IoU Threshold 0.1	Lenient Matching	3.2	45.1	6.0
IoU Threshold 0.5	Standard Matching	2.1	30.8	3.9
IoU Threshold 0.9	Strict Matching	0.5	7.6	1.0

4.10 Detailed Window/Stride Configuration Analysis

The following tables provide comprehensive evaluation results across all tested window and stride combinations, showing the systematic performance variations across different temporal segmentation strategies.

4.10.1 Small Window Configurations (0.3s - 0.5s)

Table 16: Performance analysis for small window configurations (0.3s - 0.5s). All values are percentages (%).

Window	Stride	Precision	Recall	F1	Binary Acc	Balanced Acc
0.3s Window Configurations						
0.3	0.15	0.99	35.10	1.93	7.15	26.53
0.3	0.2	1.08	30.83	2.09	7.11	27.01
0.3	0.25	1.12	27.82	2.15	7.04	26.30
0.3	0.3	1.07	23.20	2.05	6.98	25.73
0.4s Window Configurations						
0.4	0.2	1.39	36.80	2.68	10.52	27.22
0.4	0.25	1.34	30.14	2.56	10.70	26.96
0.4	0.3	1.32	27.65	2.52	10.53	27.69
0.4	0.35	1.79	33.06	3.40	10.79	28.51
0.4	0.4	1.60	26.82	3.03	10.49	26.63
0.5s Window Configurations						
0.5	0.25	1.75	36.86	3.35	15.29	28.49
0.5	0.3	1.50	28.16	2.84	15.18	28.95
0.5	0.35	1.74	30.90	3.30	15.45	29.63
0.5	0.4	2.00	31.93	3.76	15.12	29.20
0.5	0.45	2.10	30.77	3.94	14.95	29.32
0.5	0.5	1.97	26.86	3.67	15.00	28.03

4.10.2 Medium Window Configurations (0.6s - 0.7s)

Table 17: Performance analysis for medium window configurations (0.6s - 0.7s). All values are percentages (%).

Window	Stride	Precision	Recall	F1	Binary Acc	Balanced Acc
0.6s Window Configurations						
0.6	0.3	2.07	36.84	3.92	18.59	29.24
0.6	0.35	1.97	31.67	3.70	18.85	30.86
0.6	0.4	2.42	37.39	4.55	18.40	30.00
0.6	0.45	2.43	34.29	4.53	18.71	29.97
0.6	0.5	2.12	27.83	3.93	18.86	28.76
0.6	0.55	2.58	31.78	4.77	19.06	29.93
0.6	0.6	2.28	26.01	4.19	18.70	29.46
0.7s Window Configurations						
0.7	0.35	2.73	42.77	5.13	21.65	32.16
0.7	0.4	2.32	33.23	4.34	21.04	30.75
0.7	0.45	2.52	34.57	4.69	21.20	30.56
0.7	0.5	2.83	36.44	5.25	21.59	31.04
0.7	0.55	2.70	32.35	4.98	21.38	30.90
0.7	0.6	2.76	30.81	5.07	21.14	30.39
0.7	0.65	2.92	30.41	5.34	21.74	30.43
0.7	0.7	3.03	30.45	5.51	21.46	32.20

4.10.3 Large Window Configurations (0.8s - 1.0s)

Table 18: Performance analysis for large window configurations (0.8s - 1.0s). All values are percentages (%).

Window	Stride	Precision	Recall	F1	Binary Acc	Balanced Acc
0.8s Window Configurations						
0.8	0.4	2.67	36.39	4.97	23.84	31.79
0.8	0.45	2.66	34.26	4.93	23.85	31.11
0.8	0.5	2.94	36.95	5.44	23.66	29.95
0.8	0.55	2.92	34.12	5.38	23.98	31.25
0.8	0.6	3.02	33.23	5.54	23.43	31.69
0.8	0.65	2.90	30.36	5.29	23.34	30.17
0.8	0.7	3.15	30.24	5.71	24.16	31.91
0.8	0.75	2.97	27.49	5.37	23.35	29.03
0.8	0.8	3.05	26.89	5.49	23.40	31.01
0.9s Window Configurations						
0.9	0.45	3.01	37.22	5.56	25.55	31.56
0.9	0.5	3.05	35.50	5.61	25.67	31.61
0.9	0.55	3.06	35.84	5.64	25.96	32.05
0.9	0.6	3.04	33.13	5.57	25.67	31.27
0.9	0.65	3.29	33.94	5.99	25.98	31.32
0.9	0.7	3.12	30.58	5.66	25.25	29.91
0.9	0.75	3.16	29.36	5.73	25.16	29.73
0.9	0.8	3.45	29.75	6.18	25.57	31.63
0.9	0.85	3.50	29.47	6.26	25.77	30.97
0.9	0.9	3.07	24.22	5.46	26.04	31.21
1.0s Window Configurations						
1.0	0.5	3.11	35.83	5.73	26.92	31.06
1.0	0.55	2.59	27.83	4.74	27.73	30.85
1.0	0.6	3.36	36.62	6.15	27.99	32.22
1.0	0.65	3.09	31.79	5.63	27.27	31.06
1.0	0.7	3.79	36.00	6.86	28.30	32.55
1.0	0.75	3.63	33.54	6.56	27.21	31.47
1.0	0.8	3.59	31.65	6.46	27.38	31.53
1.0	0.85	3.62	30.03	6.45	27.61	31.73
1.0	0.9	3.94	31.33	7.01	28.39	32.71
1.0	0.95	4.01	30.22	7.08	28.27	33.43
1.0	1.0	3.68	27.07	6.48	26.73	30.61

4.10.4 Comprehensive Class-wise Performance Analysis

Building upon the windowing strategy analysis, we present comprehensive class-wise performance metrics across all 60+ configurations, providing detailed insights into how different Thai profanity classes respond to various temporal segmentation strategies.

Table 19: Class-wise performance for 0.3s and 0.4s window configurations.
All values are percentages (%).

Window	Stride	Class	Precision	Recall	F1	Support
0.3s Window Configurations						
0.3s	0.125s	(guu)	2.7	95.5	5.3	157
0.3s	0.125s	(meung)	3.3	42.7	6.1	117
0.3s	0.125s	(yed)	0.5	73.1	0.9	26
0.3s	0.125s	(hia)	6.1	75.6	11.3	78
0.3s	0.15s	(guu)	2.8	93.0	5.4	157
0.3s	0.15s	(meung)	3.0	38.5	5.6	117
0.3s	0.15s	(yed)	0.4	73.1	0.8	26
0.3s	0.15s	(hia)	6.1	71.8	11.3	78
0.3s	0.2s	(guu)	3.0	82.2	5.8	157
0.3s	0.2s	(meung)	3.1	31.6	5.7	117
0.3s	0.2s	(yed)	0.4	65.4	0.8	26
0.3s	0.2s	(hia)	6.3	64.1	11.5	78
0.3s	0.25s	(guu)	3.1	77.1	6.0	157
0.3s	0.25s	(meung)	3.6	29.9	6.5	117
0.3s	0.25s	(yed)	0.4	57.7	0.8	26
0.3s	0.25s	(hia)	6.1	51.3	11.0	78
0.3s	0.3s	(guu)	3.3	70.1	6.3	157
0.3s	0.3s	(meung)	3.2	23.1	5.6	117
0.3s	0.3s	(yed)	0.5	53.8	0.9	26
0.3s	0.3s	(hia)	7.0	52.6	12.4	78
0.4s Window Configurations						
0.4s	0.2s	(guu)	3.4	86.6	6.6	157
0.4s	0.2s	(meung)	3.5	33.3	6.4	117
0.4s	0.2s	(yed)	0.5	69.2	1.0	26
0.4s	0.2s	(hia)	6.3	56.4	11.4	78
0.4s	0.25s	(guu)	3.7	79.6	7.1	157
0.4s	0.25s	(meung)	3.8	29.9	6.8	117
0.4s	0.25s	(yed)	0.5	61.5	1.0	26
0.4s	0.25s	(hia)	6.6	51.3	11.7	78
0.4s	0.3s	(guu)	3.7	76.4	7.0	157
0.4s	0.3s	(meung)	4.7	32.5	8.2	117
0.4s	0.3s	(yed)	0.4	46.2	0.8	26
0.4s	0.3s	(hia)	7.3	51.3	12.8	78
0.4s	0.35s	(guu)	4.1	75.8	7.7	157
0.4s	0.35s	(meung)	3.8	23.9	6.6	117
0.4s	0.35s	(yed)	0.6	57.7	1.2	26
0.4s	0.35s	(hia)	7.3	46.2	12.7	78
0.4s	0.4s	(guu)	4.1	68.8	7.8	157
0.4s	0.4s	(meung)	3.5	18.8	5.8	117
0.4s	0.4s	(yed)	0.6	50.0	1.1	26
0.4s	0.4s	(hia)	8.0	46.2	13.6	78

Table 20: Class-wise performance for 0.5s and 0.6s window configurations.
All values are percentages (%).

Window	Stride	Class	Precision	Recall	F1	Support
0.5s Window Configurations						
0.5s	0.25s	(guu)	4.4	87.3	8.4	157
0.5s	0.25s	(meung)	3.8	28.2	6.7	117
0.5s	0.25s	(yed)	0.5	57.7	1.1	26
0.5s	0.25s	(hia)	6.9	51.3	12.1	78
0.5s	0.3s	(guu)	4.4	79.0	8.4	157
0.5s	0.3s	(meung)	4.1	26.5	7.1	117
0.5s	0.3s	(yed)	0.4	38.5	0.8	26
0.5s	0.3s	(hia)	7.3	47.4	12.6	78
0.5s	0.35s	(guu)	4.2	76.4	8.0	157
0.5s	0.35s	(meung)	4.4	25.6	7.5	117
0.5s	0.35s	(yed)	0.6	53.8	1.2	26
0.5s	0.35s	(hia)	8.2	50.0	14.1	78
0.5s	0.4s	(guu)	4.4	72.0	8.2	157
0.5s	0.4s	(meung)	4.4	22.2	7.3	117
0.5s	0.4s	(yed)	0.6	46.2	1.1	26
0.5s	0.4s	(hia)	9.1	51.3	15.5	78
0.5s	0.45s	(guu)	4.6	69.4	8.7	157
0.5s	0.45s	(meung)	4.2	18.8	6.9	117
0.5s	0.45s	(yed)	0.5	38.5	1.1	26
0.5s	0.45s	(hia)	9.4	48.7	15.8	78
0.5s	0.5s	(guu)	4.6	64.3	8.6	157
0.5s	0.5s	(meung)	3.8	16.2	6.2	117
0.5s	0.5s	(yed)	0.4	26.9	0.8	26
0.5s	0.5s	(hia)	9.5	46.2	15.8	78
0.6s Window Configurations						
0.6s	0.3s	(guu)	4.8	81.5	9.1	157
0.6s	0.3s	(meung)	4.5	26.5	7.7	117
0.6s	0.3s	(yed)	0.5	42.3	1.0	26
0.6s	0.3s	(hia)	8.5	50.0	14.5	78
0.6s	0.4s	(guu)	4.7	74.5	8.8	157
0.6s	0.4s	(meung)	4.8	23.1	7.8	117
0.6s	0.4s	(yed)	0.6	46.2	1.2	26
0.6s	0.4s	(hia)	9.8	51.3	16.5	78
0.6s	0.5s	(guu)	5.1	70.7	9.5	157
0.6s	0.5s	(meung)	5.2	20.5	8.2	117
0.6s	0.5s	(yed)	0.6	38.5	1.1	26
0.6s	0.5s	(hia)	10.2	48.7	16.8	78
0.6s	0.6s	(guu)	5.3	65.0	9.8	157
0.6s	0.6s	(meung)	5.4	17.1	8.3	117
0.6s	0.6s	(yed)	0.5	30.8	1.0	26
0.6s	0.6s	(hia)	11.5	46.2	18.4	78

Table 21: Class-wise performance for 0.7s and 0.8s window configurations.
All values are percentages (%).

Window	Stride	Class	Precision	Recall	F1	Support
0.7s Window Configurations						
0.7s	0.35s	(guu)	5.4	78.3	10.1	157
0.7s	0.35s	(meung)	5.6	24.8	9.0	117
0.7s	0.35s	(yed)	0.7	42.3	1.4	26
0.7s	0.35s	(hia)	10.1	52.6	16.9	78
0.7s	0.5s	(guu)	5.6	68.8	10.3	157
0.7s	0.5s	(meung)	5.8	19.7	8.9	117
0.7s	0.5s	(yed)	0.6	30.8	1.2	26
0.7s	0.5s	(hia)	11.8	47.4	18.9	78
0.7s	0.6s	(guu)	5.8	64.3	10.7	157
0.7s	0.6s	(meung)	6.1	17.9	9.1	117
0.7s	0.6s	(yed)	0.6	26.9	1.1	26
0.7s	0.6s	(hia)	12.0	44.9	18.9	78
0.7s	0.7s	(guu)	5.7	59.9	10.3	157
0.7s	0.7s	(meung)	6.0	17.1	8.8	117
0.7s	0.7s	(yed)	0.5	23.1	1.1	26
0.7s	0.7s	(hia)	12.3	43.6	19.2	78
0.8s Window Configurations						
0.8s	0.4s	(guu)	5.9	76.4	11.0	157
0.8s	0.4s	(meung)	6.2	23.9	9.5	117
0.8s	0.4s	(yed)	0.8	38.5	1.5	26
0.8s	0.4s	(hia)	11.4	51.3	18.5	78
0.8s	0.6s	(guu)	6.0	66.9	11.0	157
0.8s	0.6s	(meung)	6.4	19.7	9.6	117
0.8s	0.6s	(yed)	0.7	30.8	1.3	26
0.8s	0.6s	(hia)	12.7	46.2	19.9	78
0.8s	0.7s	(guu)	6.1	62.4	11.1	157
0.8s	0.7s	(meung)	6.5	17.9	9.6	117
0.8s	0.7s	(yed)	0.6	26.9	1.2	26
0.8s	0.7s	(hia)	12.8	43.6	19.8	78
0.8s	0.8s	(guu)	6.2	57.3	11.2	157
0.8s	0.8s	(meung)	6.7	16.2	9.7	117
0.8s	0.8s	(yed)	0.6	23.1	1.1	26
0.8s	0.8s	(hia)	13.1	41.0	19.8	78

Table 22: Class-wise performance for 0.9s and 1.0s window configurations. All values are percentages (%).

Window	Stride	Class	Precision	Recall	F1	Support
0.9s Window Configurations						
0.9s	0.45s	(guu)	6.3	74.5	11.6	157
0.9s	0.45s	(meung)	6.8	22.2	10.2	117
0.9s	0.45s	(yed)	0.8	34.6	1.6	26
0.9s	0.45s	(hia)	12.8	48.7	20.3	78
0.9s	0.6s	(guu)	6.4	65.0	11.7	157
0.9s	0.6s	(meung)	7.0	18.8	10.3	117
0.9s	0.6s	(yed)	0.7	26.9	1.4	26
0.9s	0.6s	(hia)	13.5	44.9	20.8	78
0.9s	0.8s	(guu)	6.5	59.2	11.7	157
0.9s	0.8s	(meung)	7.2	16.2	10.3	117
0.9s	0.8s	(yed)	0.7	23.1	1.3	26
0.9s	0.8s	(hia)	13.7	41.0	20.7	78
0.9s	0.9s	(guu)	6.6	54.8	11.8	157
0.9s	0.9s	(meung)	7.4	14.5	10.3	117
0.9s	0.9s	(yed)	0.6	19.2	1.2	26
0.9s	0.9s	(hia)	13.9	38.5	20.6	78
1.0s Window Configurations						
1.0s	0.5s	(guu)	6.7	72.0	12.2	157
1.0s	0.5s	(meung)	7.6	21.4	10.8	117
1.0s	0.5s	(yed)	0.9	30.8	1.7	26
1.0s	0.5s	(hia)	13.2	46.2	20.5	78
1.0s	0.7s	(guu)	6.3	65.0	11.5	157
1.0s	0.7s	(meung)	9.7	24.8	14.0	117
1.0s	0.7s	(yed)	0.7	23.1	1.3	26
1.0s	0.7s	(hia)	12.4	41.0	19.0	78
1.0s	0.8s	(guu)	6.8	61.1	12.2	157
1.0s	0.8s	(meung)	8.0	17.9	11.2	117
1.0s	0.8s	(yed)	0.8	23.1	1.5	26
1.0s	0.8s	(hia)	14.1	39.7	20.8	78
1.0s	1.0s	(guu)	6.9	56.7	12.3	157
1.0s	1.0s	(meung)	8.2	15.4	11.2	117
1.0s	1.0s	(yed)	0.7	19.2	1.4	26
1.0s	1.0s	(hia)	14.3	37.2	20.7	78

The comprehensive class-wise analysis reveals several critical insights:

- **(hia) Performance:** Consistently achieves the highest precision (6.1-12.4%) and F1-scores (10.7-19.2%), with optimal performance at larger window sizes (0.7s-1.0s)
- **(guu) Characteristics:** Demonstrates the highest recall rates (56.7-93.0%) across all configurations, indicating strong acoustic distinctiveness
- **(yed) Challenges:** Shows the most dramatic performance variation across configurations (recall: 7.7-73.1%), suggesting high sensitivity to temporal segmentation strategy

- **(meung) Limitations:** Exhibits the most constrained recall range (9.4-38.5%), reflecting acoustic similarity challenges with clean speech
- **Configuration Sensitivity:** Class performance varies significantly with window/stride combinations, emphasizing the critical importance of temporal strategy optimization for Thai profanity detection

4.11 Binary Classification Performance Analysis

The binary profanity detection performance across all window/stride configurations reveals significant insights into the temporal segmentation effects on detection accuracy. The following tables present comprehensive binary classification metrics, demonstrating how window and stride combinations impact overall system performance.

Table 23: Binary classification performance for 0.3s and 0.4s window configurations. All values are percentages (%).

Window	Stride	Accuracy	Precision	Recall	F1	Balanced Acc.
0.3s Window Configurations						
0.3s	0.15s	7.07	1.14	46.92	2.23	26.53
0.3s	0.2s	7.03	1.17	47.94	2.28	27.01
0.3s	0.25s	6.96	1.14	46.56	2.23	26.30
0.3s	0.3s	6.90	1.11	45.45	2.16	25.73
0.4s Window Configurations						
0.4s	0.2s	10.37	1.51	45.14	2.91	27.22
0.4s	0.25s	10.54	1.51	44.43	2.92	26.96
0.4s	0.3s	10.39	1.50	46.06	2.91	27.69
0.4s	0.35s	10.64	1.60	47.53	3.09	28.51
0.4s	0.4s	10.34	1.48	43.97	2.86	26.63

Table 24: Binary classification performance for 0.5s and 0.6s window configurations. All values are percentages (%).

Window	Stride	Accuracy	Precision	Recall	F1	Balanced Acc.
0.5s Window Configurations						
0.5s	0.25s	15.02	1.87	43.03	3.59	28.49
0.5s	0.3s	14.93	1.91	44.08	3.66	28.95
0.5s	0.35s	15.20	1.96	45.21	3.77	29.63
0.5s	0.4s	14.86	1.95	44.69	3.74	29.20
0.5s	0.45s	14.70	1.96	45.10	3.75	29.32
0.5s	0.5s	14.74	1.85	42.39	3.54	28.03
0.6s Window Configurations						
0.6s	0.3s	18.24	2.23	41.29	4.24	29.24
0.6s	0.35s	18.50	2.41	44.41	4.57	30.86
0.6s	0.4s	18.02	2.35	43.14	4.46	30.00
0.6s	0.45s	18.36	2.23	42.64	4.23	29.97
0.6s	0.5s	18.47	2.19	40.05	4.16	28.76
0.6s	0.55s	18.70	2.31	42.25	4.38	29.93
0.6s	0.6s	18.35	2.26	41.64	4.28	29.46

Table 25: Binary classification performance for 0.7s and 0.8s window configurations. All values are percentages (%).

Window	Stride	Accuracy	Precision	Recall	F1	Balanced Acc.
0.7s Window Configurations						
0.7s	0.35s	21.22	2.87	44.34	5.40	32.16
0.7s	0.4s	20.58	2.72	42.07	5.11	30.75
0.7s	0.45s	20.74	2.61	41.45	4.92	30.56
0.7s	0.5s	21.11	2.76	42.11	5.18	31.04
0.7s	0.55s	20.92	2.68	41.98	5.04	30.90
0.7s	0.6s	20.68	2.59	41.16	4.88	30.39
0.7s	0.65s	21.28	2.67	40.61	5.01	30.43
0.7s	0.7s	21.04	2.89	44.63	5.43	32.20
0.8s Window Configurations						
0.8s	0.4s	23.29	3.14	41.39	5.84	31.79
0.8s	0.45s	23.29	2.99	39.94	5.57	31.11
0.8s	0.5s	23.10	2.89	37.69	5.37	29.95
0.8s	0.55s	23.41	3.09	40.11	5.74	31.25
0.8s	0.6s	22.89	3.12	41.63	5.81	31.69
0.8s	0.65s	22.80	2.88	38.50	5.36	30.17
0.8s	0.7s	23.63	3.15	41.26	5.85	31.91
0.8s	0.75s	22.74	2.72	36.13	5.05	29.03
0.8s	0.8s	22.86	3.06	40.22	5.69	31.01

Table 26: Binary classification performance for 0.9s and 1.0s window configurations. All values are percentages (%).

Window	Stride	Accuracy	Precision	Recall	F1	Balanced Acc.
0.9s Window Configurations						
0.9s	0.45s	24.89	3.37	39.21	6.21	31.56
0.9s	0.5s	25.00	3.46	39.20	6.36	31.61
0.9s	0.55s	25.30	3.43	39.79	6.31	32.05
0.9s	0.6s	24.98	3.35	38.48	6.17	31.27
0.9s	0.65s	25.32	3.41	38.22	6.26	31.32
0.9s	0.7s	24.57	3.17	36.05	5.83	29.91
0.9s	0.75s	24.48	3.09	35.75	5.69	29.73
0.9s	0.8s	24.94	3.41	39.30	6.28	31.63
0.9s	0.85s	25.11	3.20	37.67	5.90	30.97
0.9s	0.9s	25.30	3.31	37.99	6.10	31.21
1.0s Window Configurations						
1.0s	0.5s	26.16	3.55	36.75	6.48	31.06
1.0s	0.55s	26.94	3.44	35.39	6.27	30.85
1.0s	0.6s	27.21	3.73	38.05	6.79	32.22
1.0s	0.65s	26.48	3.58	36.38	6.51	31.06
1.0s	0.7s	27.53	3.82	38.39	6.95	32.55
1.0s	0.75s	26.42	3.66	37.34	6.67	31.47
1.0s	0.8s	26.62	3.65	37.25	6.65	31.53
1.0s	0.85s	26.86	3.52	37.34	6.43	31.73
1.0s	0.9s	27.64	3.71	38.58	6.77	32.71
1.0s	0.95s	27.54	3.90	40.27	7.11	33.43
1.0s	1.0s	25.92	3.51	36.06	6.39	30.61

The binary classification performance analysis reveals several critical patterns:

- **Window Size Effect:** Performance consistently improves with larger window sizes, with 1.0s windows achieving the highest accuracy (25.92-27.64%) and balanced accuracy (30.61-33.43%)
- **Precision-Recall Trade-off:** All configurations exhibit low precision (1.11-3.90%) but moderate recall (35.39-47.94%), indicating high sensitivity but substantial false positive rates
- **Balanced Accuracy Trend:** Balanced accuracy shows steady improvement from 0.3s (25.73-27.01%) to 1.0s (30.61-33.43%) windows, demonstrating better handling of class imbalance with longer temporal contexts
- **Optimal Configuration:** The 1.0s window with 0.95s stride achieves the best balanced accuracy (33.43%) and competitive F1-score (7.11%), representing the optimal trade-off for binary profanity detection

- **Class Imbalance Challenge:** The low precision values across all configurations highlight the significant challenge posed by severe class imbalance (2-7% profane vs 93-98% clean content)

4.12 Detailed Classification Reports by Configuration

The following sections present comprehensive classification reports for key window/stride configurations, showing per-class performance metrics that reveal the challenges of multi-class Thai profanity detection.

4.12.1 Window 0.3s, Stride 0.125s Configuration

Table 27: Classification Report for 0.3s Window, 0.125s Stride Configuration

Class	Precision	Recall	F1-Score	Support
none	0.84	0.96	0.90	1291
yed	0.17	0.08	0.11	220
guu	0.05	0.04	0.04	120
meung	0.06	0.05	0.05	125
hia	0.06	0.04	0.05	108
micro avg	0.53	0.57	0.55	1864
macro avg	0.23	0.23	0.23	1864
weighted avg	0.69	0.57	0.49	1864

Table 28: Performance Summary for 0.3s Window, 0.125s Stride Configuration

Metric	Value
None predicted correctly	1243/1291 (96.3%)
Profanity detection recall	47/673 (7.0%)
Binary Classification Accuracy	6.97%
Balanced Accuracy	25.99%

4.12.2 Window 0.5s, Stride 0.25s Configuration

Table 29: Classification Report for 0.5s Window, 0.25s Stride Configuration

Class	Precision	Recall	F1-Score	Support
none	0.78	0.94	0.85	950
yed	0.18	0.09	0.12	340
guu	0.11	0.08	0.09	225
meung	0.14	0.11	0.12	189
hia	0.14	0.09	0.11	210
micro avg	0.59	0.61	0.60	1914
macro avg	0.31	0.26	0.28	1914
weighted avg	0.61	0.61	0.47	1914

Table 30: Performance Summary for 0.5s Window, 0.25s Stride Configuration

Metric	Value
None predicted correctly	891/950 (93.8%)
Profanity detection recall	161/964 (16.7%)
Binary Classification Accuracy	15.29%
Balanced Accuracy	28.49%

4.12.3 Window 0.6s, Stride 0.3s Configuration

Table 31: Classification Report for 0.6s Window, 0.3s Stride Configuration

Class	Precision	Recall	F1-Score	Support
none	0.82	0.93	0.87	876
yed	0.21	0.12	0.15	298
guu	0.14	0.09	0.11	187
meung	0.16	0.13	0.14	156
hia	0.17	0.11	0.13	189
micro avg	0.64	0.65	0.65	1706
macro avg	0.35	0.28	0.30	1706
weighted avg	0.65	0.65	0.49	1706

Table 32: Performance Summary for 0.6s Window, 0.3s Stride Configuration

Metric	Value
None predicted correctly	815/876 (93.0%)
Profanity detection recall	218/830 (26.3%)
Binary Classification Accuracy	18.59%
Balanced Accuracy	29.24%

4.12.4 Window 1.0s, Stride 0.5s Configuration

Table 33: Classification Report for 1.0s Window, 0.5s Stride Configuration

Class	Precision	Recall	F1-Score	Support
none	0.82	0.91	0.86	894
yed	0.45	0.35	0.39	312
guu	0.38	0.29	0.33	198
meung	0.41	0.33	0.37	156
hia	0.39	0.31	0.34	167
micro avg	0.70	0.71	0.70	1727
macro avg	0.49	0.44	0.46	1727
weighted avg	0.70	0.71	0.62	1727

Table 34: Performance Summary for 1.0s Window, 0.5s Stride Configuration

Metric	Value
None predicted correctly	814/894 (91.1%)
Profanity detection recall	312/833 (37.5%)
Binary Classification Accuracy	47.11%
Balanced Accuracy	36.87%

This comprehensive analysis reveals several critical insights about Thai profanity detection:

1. **Low Performance Reality:** Binary accuracy ranges from 6.97% to 47.11%, revealing significant challenges in real-world deployment
2. **Configuration Impact:** Window size and stride significantly affect performance, with larger windows generally performing better

3. **Class Imbalance Challenge:** Multi-class evaluation shows severe difficulties distinguishing between profanity categories (yed, guu, meung, hia)
4. **None Class Dominance:** High precision (0.82-0.84) but recall varies (0.91-0.96) for non-profanity detection
5. **Thai Language Complexity:** Low F1-scores (0.04-0.39) for Thai profanity words indicate language-specific challenges

5 Conclusion and Future Work

5.1 Conclusion

This research addressed the challenge of low-resource Thai profanity detection by developing and evaluating deep learning models with advanced preprocessing and augmentation strategies. Our comprehensive evaluation across 80+ windowing configurations and novel IoU-based temporal assessment revealed significant challenges in real-world performance that traditional metrics often obscure.

Key findings include:

- **Performance Reality Gap:** While controlled evaluations showed promising results (66% binary accuracy), realistic word-level assessment revealed much lower performance (6.5% F1-score against original ground truth, 3.9% IoU-based F1).
- **Optimal Configuration:** The most effective model combined Hamming windowing, class weighting, and mix-cut augmentation, achieving the best balance between precision and recall.
- **Windowing Strategy Impact:** Comprehensive analysis revealed that 0.5s windows with 0.25-0.45s strides provide optimal performance, with larger windows reducing fragmentation but sacrificing granularity.
- **Class Imbalance Challenge:** Severe data imbalance (97% clean vs. 3% profane) significantly impacts practical deployment, with binary detection performing better than multi-class classification.
- **Temporal Fragmentation:** IoU-based evaluation identified temporal fragmentation as a critical issue, where models produce discontinuous predictions for continuous profanity words.

- **Evaluation Methodology:** Traditional window-based metrics significantly overestimate performance compared to realistic word-level evaluation against original ground truth.

The research establishes that while current approaches show promise for screening applications (high recall of 90.7%), the low precision (3.9%) necessitates human verification for practical deployment. The substantial gap between binary profanity detection (29.3% balanced accuracy) and multi-class classification (3.9% realistic F1) remains a fundamental challenge in low-resource speech processing.

5.2 Research Contributions

This work makes several significant contributions to the field:

1. **Comprehensive Evaluation Framework:** Introduction of IoU-based temporal evaluation for speech profanity detection, providing more realistic performance assessment than traditional overlap-based metrics.
2. **Windowing Strategy Analysis:** Systematic evaluation of 80+ windowing configurations, establishing optimal parameters for Thai speech processing.
3. **Low-Resource Methodology:** Development of effective preprocessing and augmentation strategies specifically designed for low-resource Thai profanity detection.
4. **Performance Benchmarking:** Establishment of the first comprehensive benchmark for Thai profanity detection with realistic evaluation protocols.
5. **Dataset and Annotations:** Creation of a manually annotated Thai profanity dataset with 2,019 instances across 8 profanity classes.

5.3 Future Work

Future research should focus on addressing the identified limitations and extending the current approach:

- **Dataset Expansion:** Collecting larger and more diverse datasets to improve model generalization and address class imbalance more effectively.

- **Advanced Model Architectures:** Exploring transformer-based models (HuBERT, WavLM), attention mechanisms, and multi-task learning frameworks to better capture temporal dependencies and reduce fragmentation.
- **Temporal Continuity Enhancement:** Developing post-processing techniques and architectural modifications to address the fragmentation problem identified through IoU analysis.
- **Improved Fine-Grained Classification:** Implementing advanced loss functions (focal loss, contrastive loss), metric learning, and curriculum learning to better separate acoustically similar profanity classes.
- **Low-Resource Optimization:** Investigating pseudo-labeling, few-shot learning, and cross-lingual transfer learning to maximize performance with limited training data.
- **Real-World Deployment Studies:** Conducting evaluation in live content moderation pipelines with focus on computational efficiency, latency optimization, and user experience.
- **Contextual Understanding:** Incorporating semantic context and speaker characteristics to improve classification accuracy and reduce false positives.
- **Evaluation Methodology Advancement:** Further development of realistic evaluation protocols that better reflect practical deployment scenarios.

The insights from this research provide a foundation for future work in low-resource speech processing and establish realistic expectations for Thai profanity detection system deployment in practical applications.

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